

HW 6: Choosing Your Inference Method

For each problem below, identify the inference procedure that needs to be used to answer the question. If you are creating a confidence interval, use your calculator to get the interval and interpret it in the context of the problem. If you are running a significance test, identify the hypotheses, use your calculator to get the test statistic and the p-value, and then write a conclusion in the context of the problem.

1) At a certain high school, a health survey was given out to determine the average height of students at the school. Out of a sample of 40 students, the average height was 5'8" with a standard deviation of 7". Construct and interpret a 95% confidence interval for the true height of all students at the school.

2) A doctor wants to determine if a new drug reduces blood pressure. Each of the 8 subjects will get the new drug and a placebo. The order the subject gets the drug and placebo will be randomly assigned. They will get one for 3 weeks and then switch and get the other for 3 weeks. At the end of each period of time, their blood pressure is recorded. Is there sufficient evidence at the 5% significance level to conclude that the drug reduces blood pressure?

Subject	1	2	3	4	5	6	7	8
Placebo BP	150	160	140	130	145	148	150	154
Drug BP	145	155	140	132	138	146	140	150

3) A political analyst wants to compare the proportion of support for a candidate in two different cities. In the first city, 85 out of 200 people support the candidate, while in the second city, 120 out of 250 people support them. Is there convincing evidence at the 5% significance level that the support for the candidate is higher in city 2 than in city 1?

4) A company wants to compare the average weekly sales of two different store locations. The first store's weekly sales (in hundreds of dollars) for 8 weeks are 150, 160, 155, 170, 145, 160, 165, 158. The second store's weekly sales (in hundreds of dollars) for the same 8 weeks were: 140, 150, 155, 145, 150, 152, 148, 155. Construct and interpret a 99% confidence interval for the difference in average weekly sales.